

	Qualitative Results
Comments	1. “<i>This is incredible and has a lot of potential. Can’t wait for this to be in real time!</i>” 2. “<i>Very well thought out! Great resource to the university</i>” 3. “<i>Lots of new people to choose from here! Great work!</i>” 4. “<i>Very useful tool overall, I could see the practical usage of this work!</i>” 5. “<i>Would love an explanation for all of your methods used!!</i>”
Feedback	1. “<i>Seeing many team pitches here from interdisciplinary domains. Can we choose from say, two, settings where we may choose to work with those from a similar domain or a different one? And how so is the overall goodness score calculated?</i>” 2. “<i>A search bar would be great for this one, not just a dropdown!</i>” 3. “<i>In addition to the proposal, can we also add research interest as a user-given input? As a merging of the second and third use cases</i>” 4. “<i>can we build our own teams for a grant?</i>”

Table 2: A sample of the comments as well as feedback for improvement received from the human study.

Conclusion

To conclude, we presented the problem of building teams for funding that allows for collaboration opportunities. We then created and implemented AI methods using string, taxonomy, and more advanced contextual boosted bandits in ULTRA, and demonstrated them to be quite useful both quantitatively (where informed methods increase recommendation quality while reducing their volume) and qualitatively in real human evaluations. We showed the generality of our approach in two different settings from US and India, and discussed our experience of deploying the system.

One area to extend our work is by considering larger data sizes for both researchers and RFPs, and from more diverse sources. Furthermore, since our methods are dependent on open data about researchers as well as proposal calls, this dependency can be both a source of strength and weakness. Data has the potential to encourage teaming without human bias but can also lead to inferior recommendation if the data is obsolete. Similarly, any feedback on the success of recommendation is only possible when a proposal has been won, and this data is usually not available or quite delayed (months or years after recommendation) to be useful. Considering a longer time frame for recommendation with proposal success data could lead to better results. Another area is to further enhance our goodness score with more metrics

Method	Average Quality	Average Volume
$M0$	0.0896 ± 0.0006	10
$M1$	0.4218 ± 0.0011	8
$M2$	0.4292 ± 0.0017	7
$M3$	0.5835 ± 0.0203	1

Table 3: Average quality (G) and volume of teams ($\#T$) shown *per* researcher (r_j) at IIT-R. This was done for each method M_i , across 100 RFPs and 46 researchers. For average quality, we report the mean and standard deviation, denoted as $\text{mean} \pm \text{STD}$.

(e.g., considering the relevance of a researcher’s previous projects to current RFPs and the number of ongoing projects a researcher is engaged in).

Our work inspires several interesting future extensions: considering a variety of domain knowledge including but not limited to fairness constraints, teaming constraints, domain constraints, etc. and developing a knowledge-driven learning system that can both exploit both the data and such knowledge remains an exciting future direction. A second direction would be to develop methods that would allow for interactive teaming where the system could not only present the recommendations but explain why and allow for human inputs to be used for refining the learned models. In addition, the scale of our survey could also be improved to ask about more aspects of the teams from participants: diversity and the connection strength between members, etc. A final and important direction is to scale this to different collaborative settings including but not limited to: healthcare, finance, law/legal, mental health, and educational support.

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